

```
# Sandstorm_Mod_Manager
# Created for Insurgency Sandstorm 1.21 by Joanna Wick
# Version: 1.0
# Date: 2026-06-20
```

This is where the game stores all information about each mod that it uses in the game. This script will also download, install and update the state.json file for any new mods you have subscribed without having to load the game.

If you want to make sure any new mods or new updates are installed just run this script before playing the game. The scan is very fast. Any updated mod or map will be downloaded and installed as well as anything new you have subscribed. You won't have to worry if the game is actually downloading anything anymore when you start the game.

The Download Progress Bar has been disabled because it slows the actual downloads to a crawl. A 1MB download could take 15-20+ minutes no matter how fast your connection. With the Download Progress disabled the downloads are as fast as your connection speed. After downloading you will have an un-zip progress indicator that will run for each download.

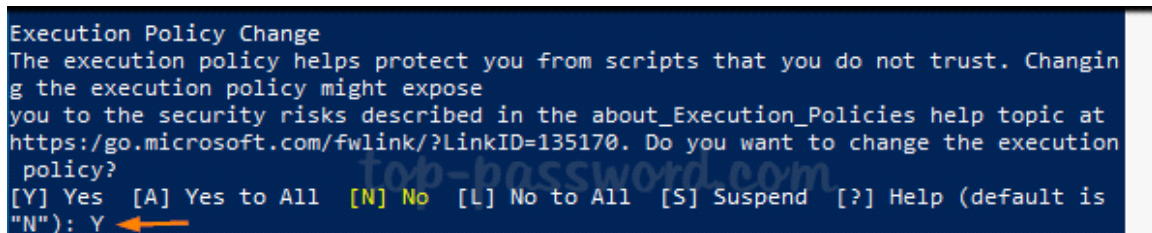
The first time you run the script it will re-download all of the mods you are subscribed and reinstall them, build a new state.json and store it's own ModList.json for use next time you run the script. I am doing this because sometimes the mod files that are actually stored do not match what the game thinks is stored.

After the first time running the script it will only download and update any files that have changed on mod.io since the last time the script was run.

If you Unsubscribe to any mod on Mod.io just run this script and it will delete the directories containing those mods so you will not need to do it manually.

First Time running Powershell Script

=====



```
Execution Policy Change
The execution policy helps protect you from scripts that you do not trust. Changing
the execution policy might expose
you to the security risks described in the about_Execution_Policies help topic at
https://go.microsoft.com/fwlink/?LinkID=135170. Do you want to change the execution
policy?
[Y] Yes [A] Yes to All [N] No [L] No to All [S] Suspend [?] Help (default is
"N"): Y
```

If you haven't run a Powershell script before or if your permissions are not defined you may see the above message then you run the script. All you need to do is enter an A or Y and the script will run without issue. If you enter N, L or S the script will NOT run and you may have a hard time getting it to execute without making policy changes. So, do not enter N, L or S.

The First thing to do is get a MOD.IO OAuth Token

=====

```
Windows PowerShell

This is your first time running Sandstorm_Mod_Manager and you need to create
an OAuth access token on mod.io so this script has access to all of your
subscribed Insurgency Sandstorm mods. For detailed information on how to do
this please read the Sandstorm_Mod_Manager_Guide.pdf file.

As this is your first time running this script all of your subscribed mods
will be re-downloaded and updated to make sure they are all up to date.
After this process only newly updated or subscribed mods will be downloaded
and updated. Do not Abort this process once started.

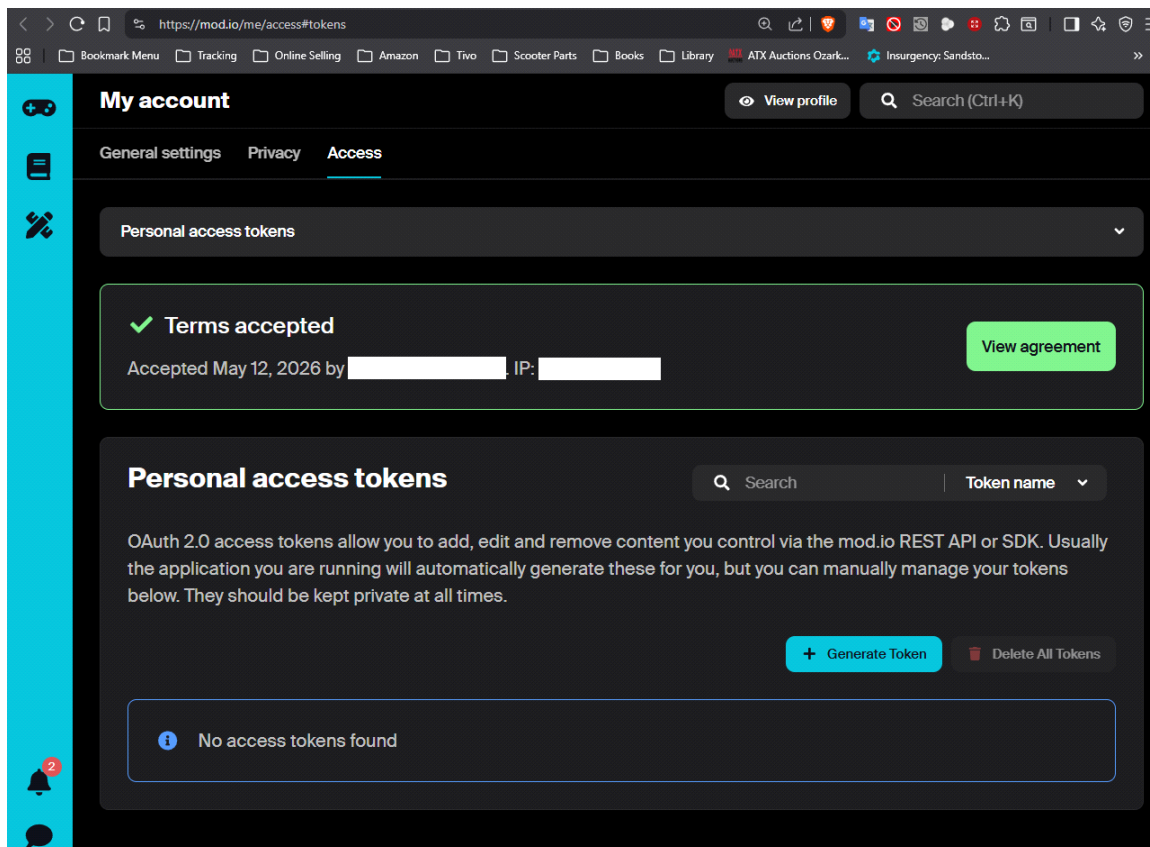
Would you like to read the Sandstorm_Mod_Manager_Guide.pdf now? [y/n]: n

Would you like to open a Web Browser to mod.io to create your OAuth token? [y/n]: n

Make sure you have your access token ready as the script will not work without one.

If you have already created and saved a token it will be shown between the Brackets [] below
OAuth token (Press ENTER to use CURRENT) []:
```

If this is your first time running the script you will see the above display asking if you want to view Sandstorm_Mod_Manager_Guide.pdf. It will allow you to open the Access page and the Personal access tokens tab on mod.io. This where you will create the OAuth Token you will need to use this script. If you click Y(es) your default web browser will open to the Access page of mod.io as shown below.



If you have never created a Personal Access Token before you will see the above page. Click on the blue + Generate Token button.

Generate personal access token

Create a personal OAuth 2.0 access token to authenticate with mod.io's REST API or SDK.

Token name *

Sandstorm

Access game *

Insurgency: Sandstorm



Description

My Mods

Permissions *



Read



Write

Expires *

Cancel

Generate

A window will open asking for some information.

1. Token Name use Sandstorm.

2. Access Game start entering Insurgency: Sandstorm. Part way through a suggestion with the full name will popup. Just click on it to populate the entry.

3. Enter anything you want for the description

4. Permissions leave as Read

Scroll down

Generate personal access token

insurgency: sandstorm

Description

My Mods

Permissions *

☒ Read

☐ Write

Expiry *

☐ 30 Days

☐ 60 Days

☐ 90 Days

☐ 6 Months

☒ 1 Year

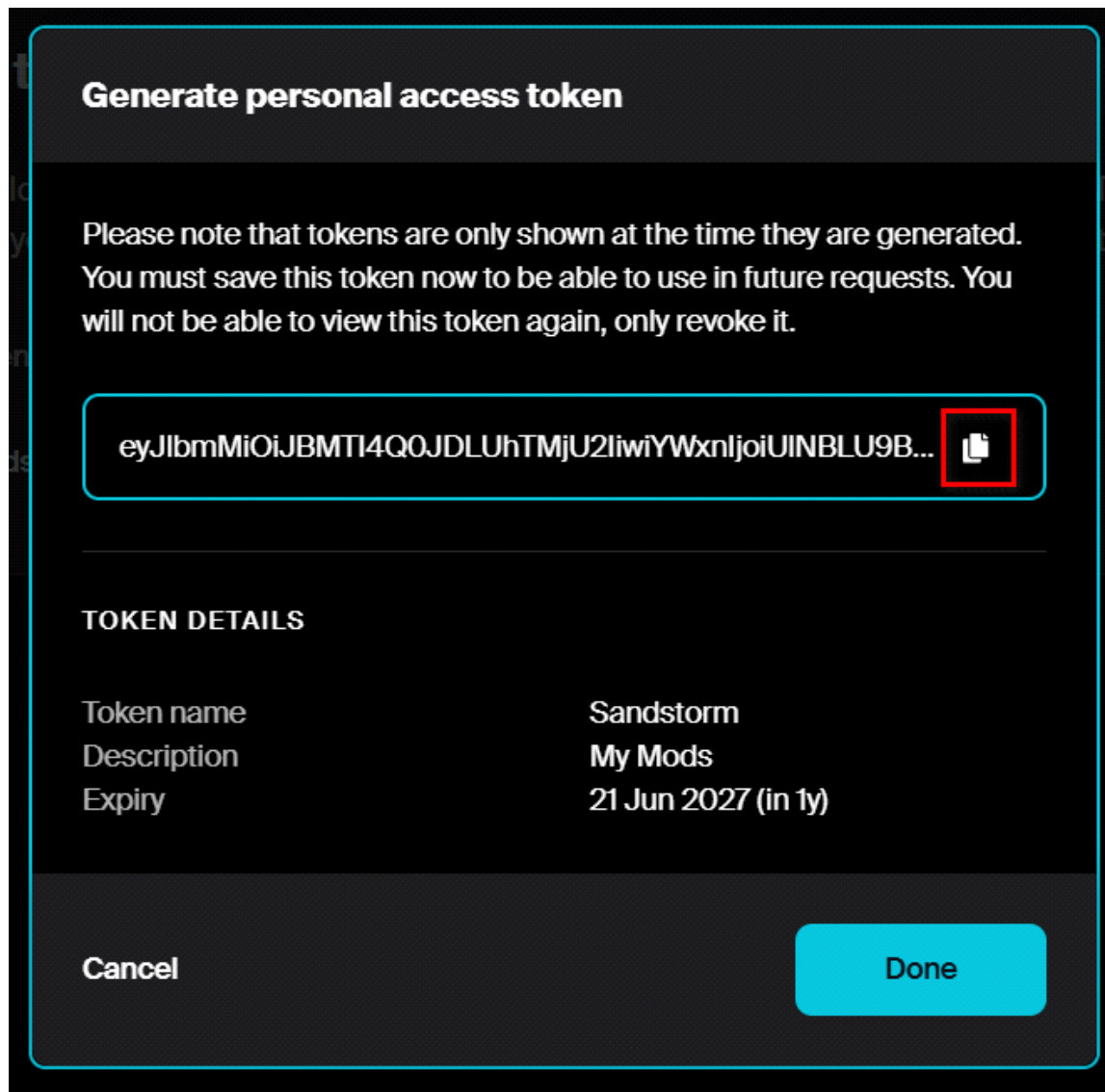
This token will expire on 21 Jun 2027

Cancel

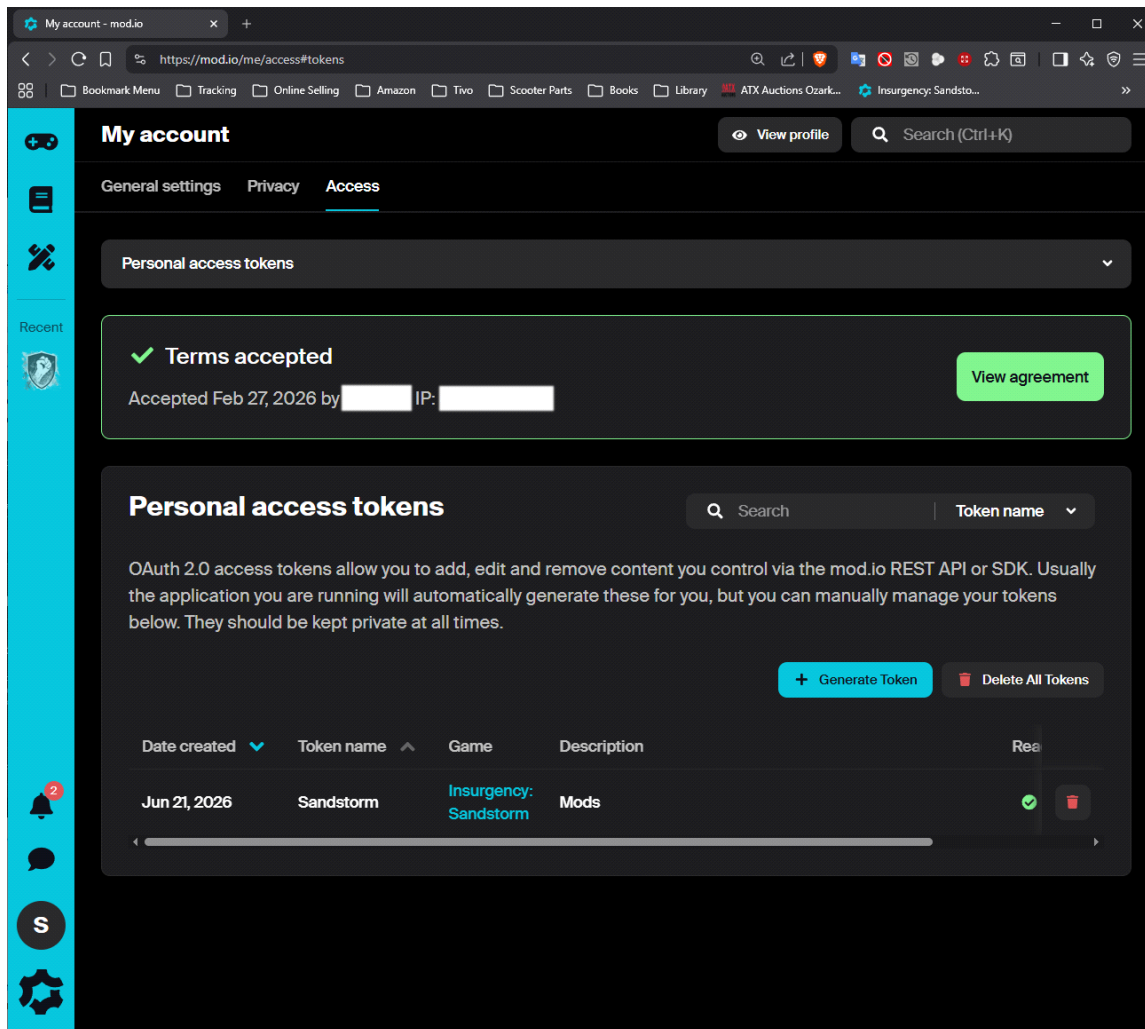
Generate

5. Expiry click on 1 year. This is how long the token will last before you need to get a new one.

Click on the blue Generate button.



6. Click on the Two Pages symbol I have highlighted with a red box. This will copy the token to your clipboard. I would suggest pasting the code into a text file to save it. Once you click on the blue Done button YOU CANNOT GET THE CODE AGAIN. If you didn't save the code you will have to create a new one and copy it. At the end of one year the Sandstorm Mod Manager will ask you to generate a new code.



When you click on the blue Done button you will see a new Token listed. You cannot access this token to get the token information. If you need the token information you will need to delete and make it again. Always SAVE the token in a text file.

```
Make sure you have your access token ready as the script will not work without one.
If you have already created and saved a token it will be shown between the Brackets [] below
OAuth token (Press ENTER to use CURRENT) []:
```

You will be asked to enter your OAuth 2 code. Just paste the code and press Enter. The code will be saved in a token.cfg file so you won't be asked again.

Sandstorm Mod Manager Menu

=====


```
Windows PowerShell

=====
Sandstorm Mod Manager
=====

1. Process Mod Subscriptions
2. Select Mods to FORCE Download and Update
3. Select Mods to Unsubscribe and Delete
4. Install/unpack mods to different location for testing purposes
   Current mod.io path: F:\SteamLibrary\steamapps\common\sandstorm\Insurgency\mods\mod.io\
5. Would you like to view Verbose Mod Information (not saved)
   Current Setting: Concise
6. Change OAuth token
   Success: The token is valid!
   Authenticated as: S[redacted]
   User ID: 5[redacted]
7. Move Mod.io Mod Directory to new Location
8. Exit

=====
Please enter your selection (1-8):
```

If your token is good and hasn't expired your Username and User ID will be displayed. If not then you will be taken to the Change OAuth screen to enter a new Personal Access Token.

1. Process Mod Subscriptions

=====

```
Windows PowerShell

=====
Downloading your Subscriptions
=====

Found 1 Subscription Page.
Found 72 subscriptions.

=====

Processing 1 of 72 Subscriptions : Warehouse
modid: 6130558
modFilename: modfile_6130558_a.zip
filesize: 639.38 MB
moddate_added: 2026-06-11 22:52:42
modVersion: 1.0.1
modURL: https://g-254.modapi.io/v1/games/254/mods/6130558/files/7839678/download

Skipped Warehouse up to date
=====
```

Mod.io only allows a maximum of 100 mods to be retrieved from your subscriptions each time. They are divided into pages. You will be shown how many pages have been found and how many mods you are subscribed. The above image shows the Verbose setting to show the modid, filename, compressed size, date added/updated, version and the direct URL used for downloading.

Windows PowerShell

```
=====
Download your Subscriptions
=====
```

Found 1 Subscription Page.

Found 72 subscriptions.

```
=====
Processing 1 of 72 Subscriptions : Warehouse
```

Skipped Warehouse up to date

```
=====
Processing 2 of 72 Subscriptions : Estates
```

Skipped Estates up to date

```
=====
Processing 3 of 72 Subscriptions : Mashtuur
```

Skipped Mashtuur up to date

```
=====
Processing 4 of 72 Subscriptions : Daybreak
```

Skipped Daybreak up to date

```
=====
Processing 5 of 72 Subscriptions : Hacienda
```

Skipped Hacienda up to date

```
=====
Processing 6 of 72 Subscriptions : Compass Radar Remake
```

Skipped Compass Radar Remake up to date

```
=====
Processing 7 of 72 Subscriptions : The Open Pit
```

Skipped The Open Pit up to date

```
=====
Processing 8 of 72 Subscriptions : NoRestrictPlus
```

Skipped NoRestrictPlus up to date

This is the regular condensed display.

```

Processing 70 of 72 Subscriptions : More Ammo Mutator [1.15]
modid: 98373
modFilename: filedata-5qlr.zip
filesize: 0.01 MB
moddate_added: 2023-10-30 19:46:51
modVersion: 0.8.4
modURL: https://g-254.modapi.io/v1/games/254/mods/98373/files/4468622/download

  Downloading More Ammo Mutator [1.15] - 98373 - filedata-5qlr.zip - 0.01 MB
  Unpacking More Ammo Mutator [1.15] - 98373 - filedata-5qlr.zip
7-Zip (a) 26.01 (x64) : Copyright (c) 1999-2026 Igor Pavlov : 2026-04-27
Scanning the drive for archives:
1 file, 12191 bytes (12 KiB)

Extracting archive: zip\filedata-5qlr.zip
--
Path = zip\filedata-5qlr.zip
Type = zip
Physical Size = 12191

Everything is Ok

Size:          48477
Compressed: 12191
Success: Archive extracted with no errors.
=====

```

When mods are being downloaded there isn't a progress bar shown. I found that a 1MB download would take 15-20 minutes to download no matter what your connection speed. I have disabled the progress bar. The file will unzip to the mods directory with a progress indicator after downloading is complete.

```

=====
Processing 64 of 72 Subscriptions : Winter
  Downloading Winter - 164381 - filedata-0yto.zip - 266.93 MB

  Download complete!
  Time Elapsed : 7.25 seconds
  Download Speed: 294.66 Mbps

  Unpacking Winter - 164381 - filedata-0yto.zip
7-Zip (a) 26.01 (x64) : Copyright (c) 1999-2026 Igor Pavlov : 2026-04-27

```

After a file has downloaded it will display the amount of time it took to download and the approximate download speed.

```

=====
Successfully Installed: 3 file(s)

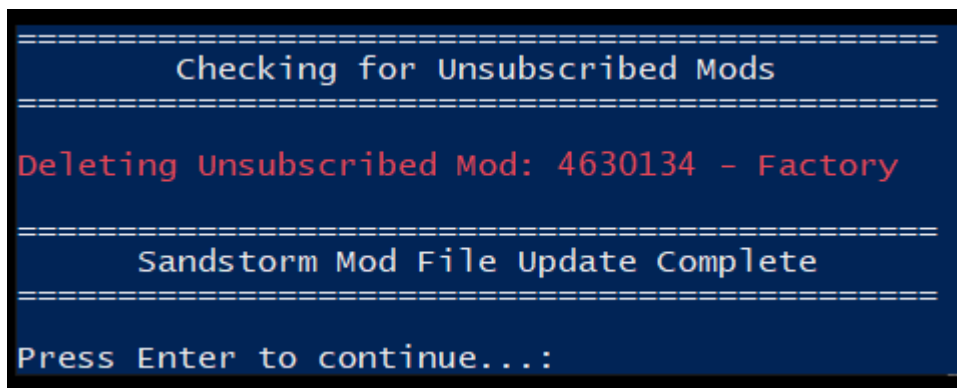
Success: Mashtuur - 5844541 - modfile_5844541.13.zip no errors.
Success: Insurgency 2014 Map Pack - 851078 - filedata-8nui.zip no errors.
Success: More Ammo Mutator [1.15] - 98373 - filedata-5qlr.zip no errors.
=====

```

A list of all successfully installed mod files is shown. Any downloads that give warnings during unzip will be listed next. The warnings can usually be ignored. Finally any downloads with errors will be listed. You should scroll up to see if they were fatal. You can always use the FORCE download option to

download those files again.

NOTE: If you see a zip file had CRC-ERRORs while unpacking it means the calculated CRC doesn't match the one in the zip file. The contents of the zip file will still be unpacked and usable by Sandstorm. Sometimes 7-Zip will claim it's a Fatal Error when in fact it isn't for Sandstorm. I have tested this on a map called Daybreak (if the map is updated the error may go away). When it is being unzipped a Fatal CRC-ERROR occurs but the files are still unpacked and the game plays the map without issue.



If there were any unsubscribed mods found the directories containing them will be deleted.

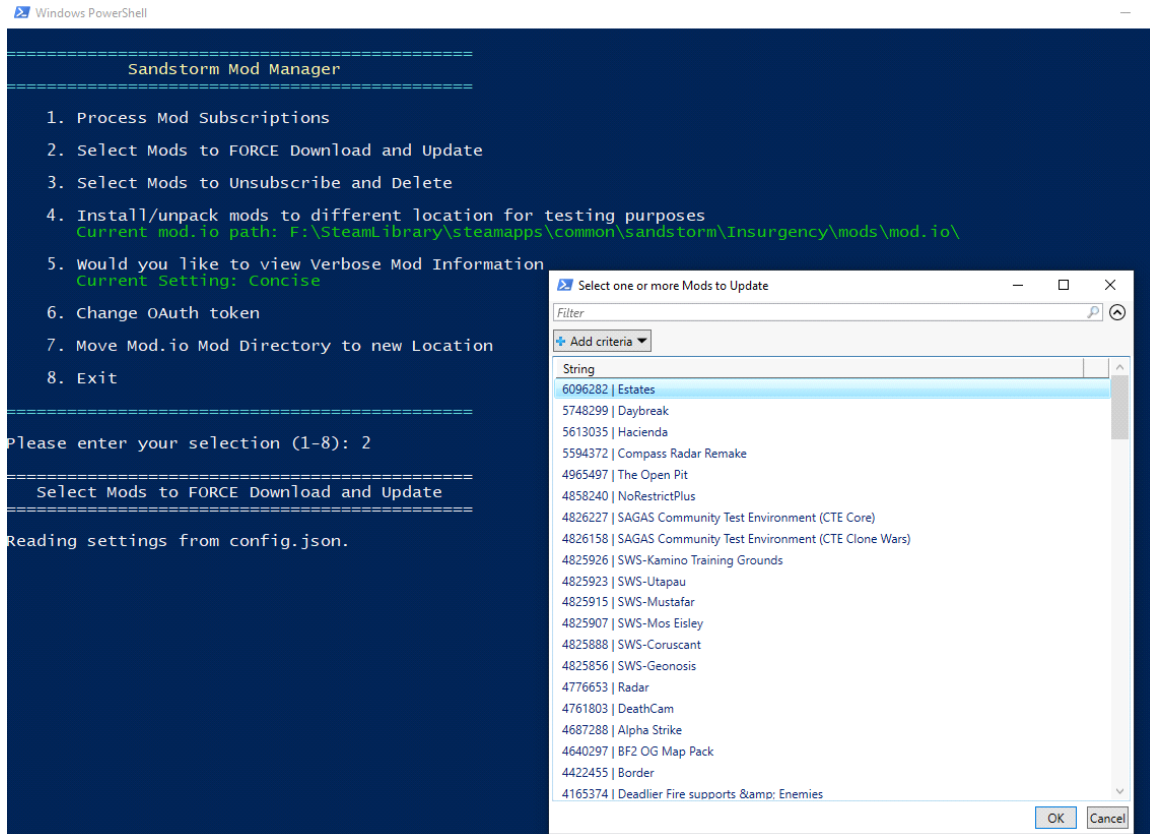
Once completed Sandstorm will have all mods updated and a new state.json file created. You can load the game and shouldn't have anything download. It could still happen if a mod was updated between the time you ran the script and loaded the game. You will still see the Downloading Server Mods... screen if the server adds new mods since the last time you joined. If on the remote chance it does stop on the Downloading server mods... screen after running this script you can exit the game to Desktop and reload and the problem should be solved. Just another one of the weird quirks with how the game incorporated the mod.io system.

Once completed Sandstorm will have all mods updated and a new state.json file created. You can load the game and shouldn't have anything download. It could still happen if a mod was updated between the time you ran the script and loaded the game. You will still see the Downloading Server Mods... screen if the server adds new mods since the last time you joined.

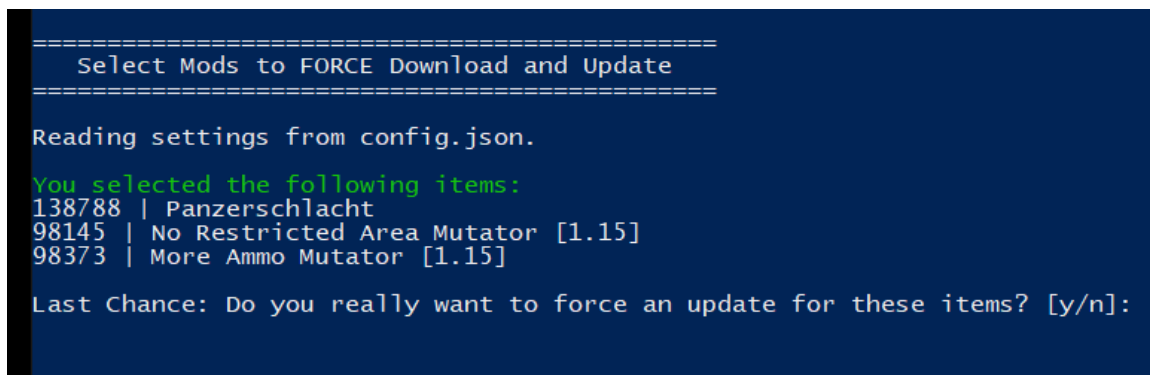
If you have subscribed to mods directly through the mod.io website you can run this script and update your mods to include them before you run the game.

2. Select Mods to FORCE Download and Update

=====



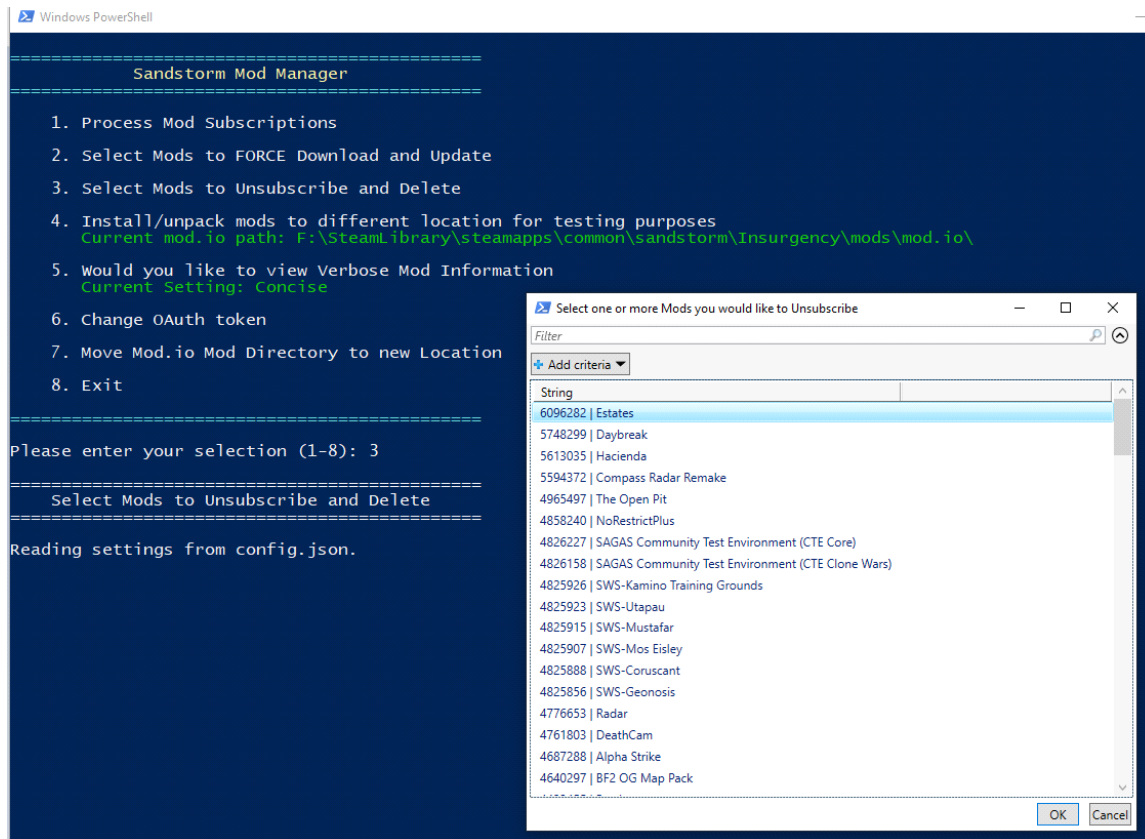
A GUI will open showing a list of all the mods you are subscribed. You can click on as many mods as you would like. The GUI uses the standard method for selecting multiple items on Windows.



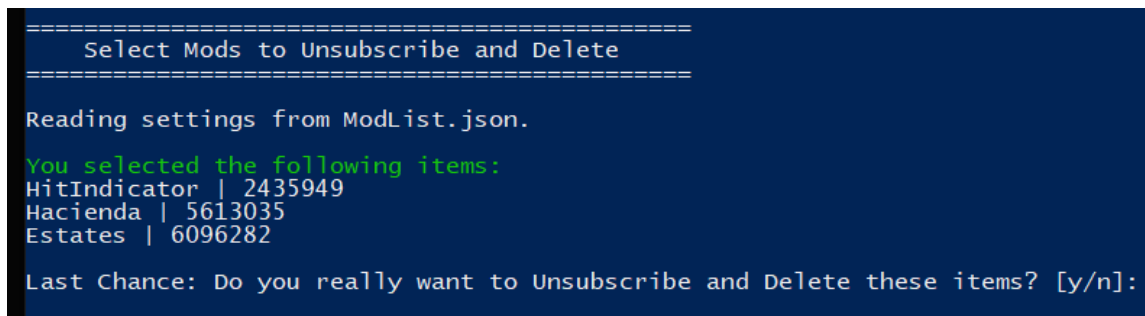
You will then be shown all of the mods you would like to re-download and ask if you really want to do it. If you enter Y(es) then the script will redownload those mods and check for any updates or new subscriptions.

3. Select Mods to Unsubscribe and Delete

=====



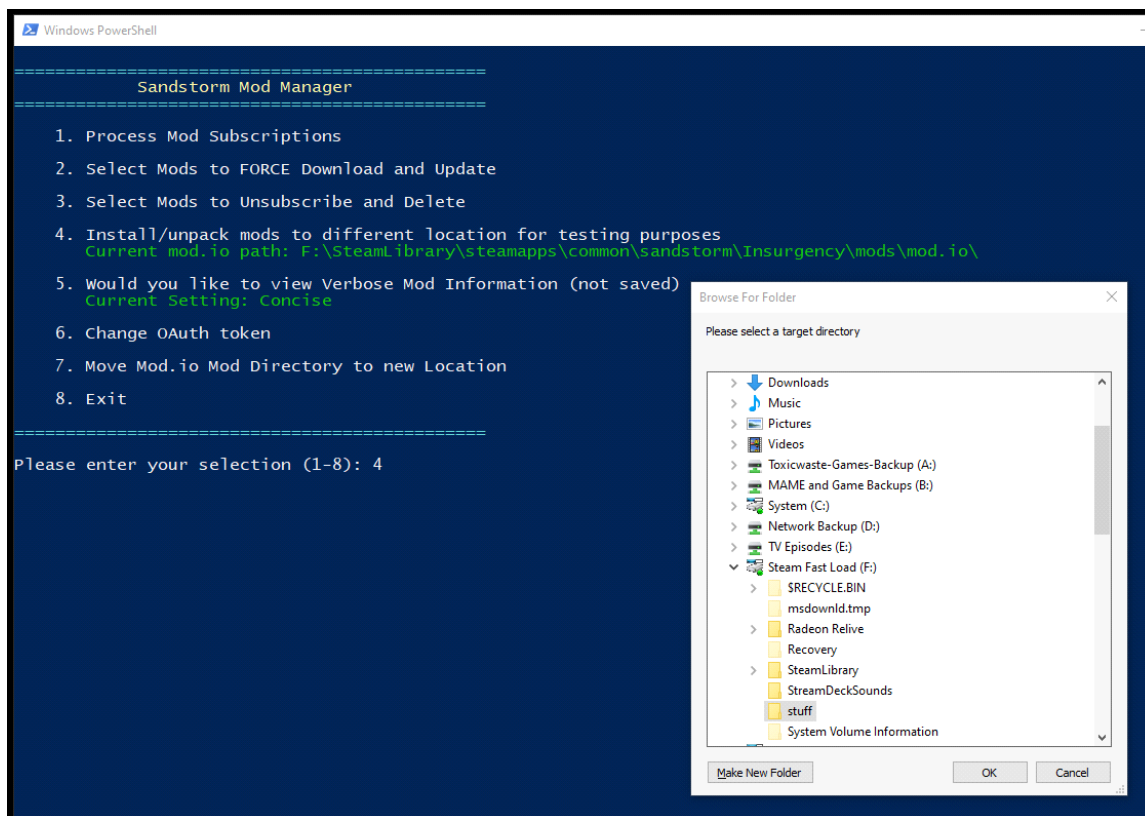
You will then be shown all of the mods you would like to Unsubscribe and ask if you really want to do it.



If you enter Y(es) then the script will Unsubscribe to those mods and Delete the directories for those mods from your computer.

4. Install/unpack mods to different location for testing purposes

=====



If you want to test download you have that option. The script will download all subscribed mods to the directory of your choice. A mod.io directory, metadata and mods sub-directories will be created. Mods are stored in mods and the rebuilt state.json in the metadata directory. This is only used for testing purposes.

```
4. Install/unpack mods to different location for testing purposes
Current mod.io path: F:\stuff\mod.io\
```

The item 4 in the menu will reflect the new location to download and store mods. Be aware that the ModList.json will be used for downloading so only new or updated mods will be stored in the test location.

I retained this mainly for my testing purposes and anyone else should use this with caution. YOU HAVE BEEN WARNED!

5. Would you like to view Verbose Mod Information

=====

Concise Display

Windows PowerShell

```
=====
Download your Subscriptions
=====
```

```
Found 1 Subscription Page.
```

```
Found 72 subscriptions.
```

```
=====
Processing 1 of 72 Subscriptions : Warehouse
```

```
Skipped Warehouse up to date
```

```
=====
Processing 2 of 72 Subscriptions : Estates
```

```
Skipped Estates up to date
```

```
=====
Processing 3 of 72 Subscriptions : Mashtuur
```

```
Skipped Mashtuur up to date
```

```
=====
Processing 4 of 72 Subscriptions : Daybreak
```

```
Skipped Daybreak up to date
```

```
=====
Processing 5 of 72 Subscriptions : Hacienda
```

```
Skipped Hacienda up to date
```

```
=====
Processing 6 of 72 Subscriptions : Compass Radar Remake
```

```
Skipped Compass Radar Remake up to date
```

```
=====
Processing 7 of 72 Subscriptions : The Open Pit
```

```
Skipped The Open Pit up to date
```

```
=====
Processing 8 of 72 Subscriptions : NoRestrictPlus
```

```
Skipped NoRestrictPlus up to date
```

Verbose display


```

Processing 70 of 72 Subscriptions : More Ammo Mutator [1.15]
modid: 98373
modFilename: filedata-5qlr.zip
filesize: 0.01 MB
moddate_added: 2023-10-30 19:46:51
modVersion: 0.8.4
modURL: https://g-254.modapi.io/v1/games/254/mods/98373/files/4468622/download

  Downloading More Ammo Mutator [1.15] - 98373 - filedata-5qlr.zip - 0.01 MB
  Unpacking More Ammo Mutator [1.15] - 98373 - filedata-5qlr.zip
7-Zip (a) 26.01 (x64) : Copyright (c) 1999-2026 Igor Pavlov : 2026-04-27
Scanning the drive for archives:
1 file, 12191 bytes (12 KiB)

Extracting archive: zip\filedata-5qlr.zip
--
Path = zip\filedata-5qlr.zip
Type = zip
Physical Size = 12191

Everything is Ok

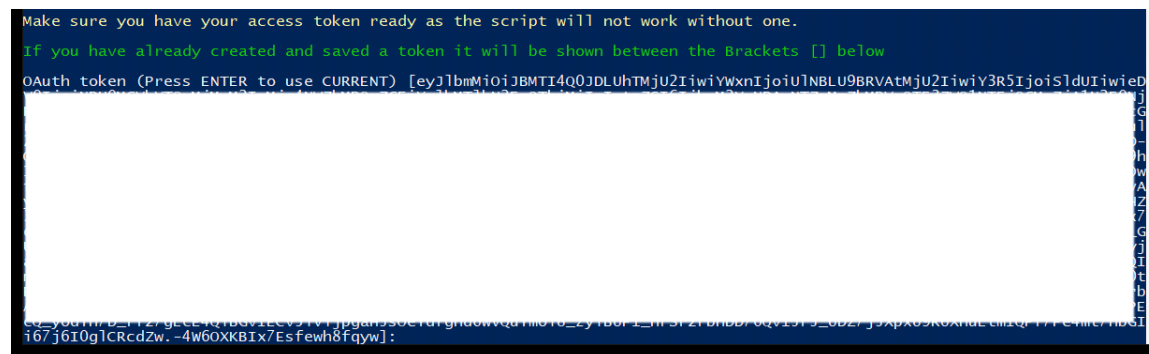
Size:      48477
Compressed: 12191
Success: Archive extracted with no errors.
=====

```

Toggle between Concise or Verbose display option when processing Subscriptions.

6. Change OAuth Token

=====

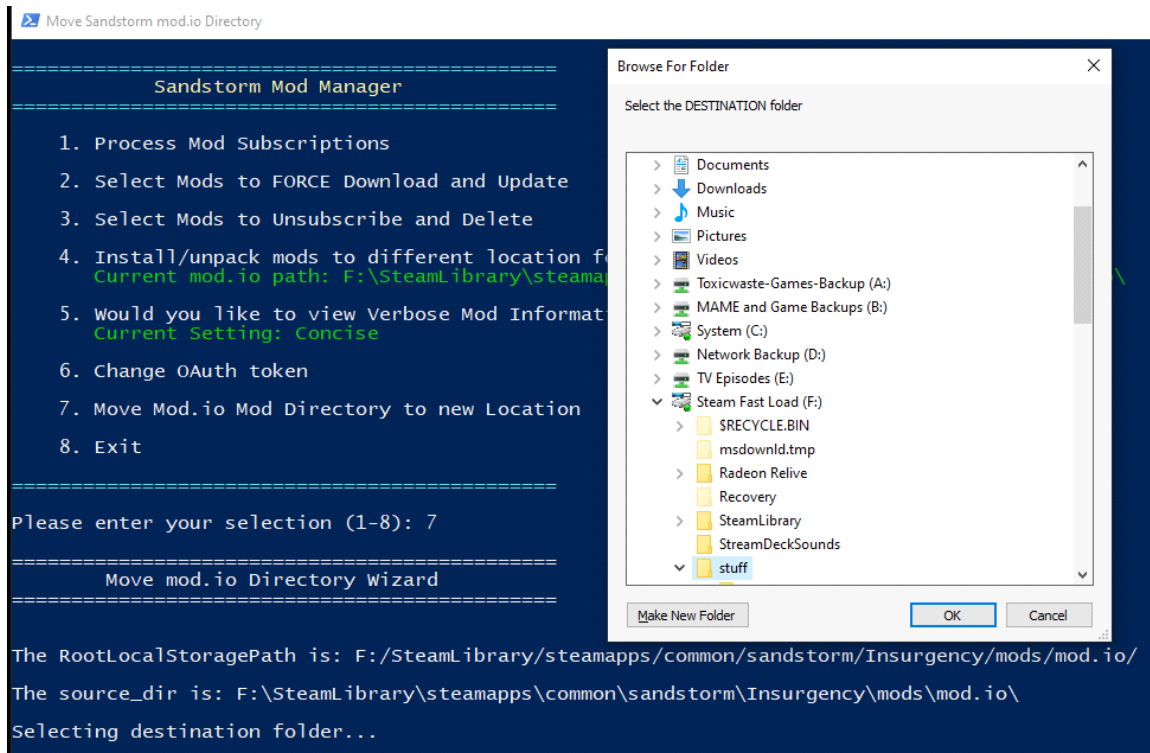


Your current Personal Access Token will be displayed as shown in the above image. Just press Enter if you want to keep the token. If you have created another token or the current token has expired you paste it here and press Enter to update. These Tokens do expire and when they do the script will notify you and take you to the Change OAuth Token page. Tokens usually last for 1 year.

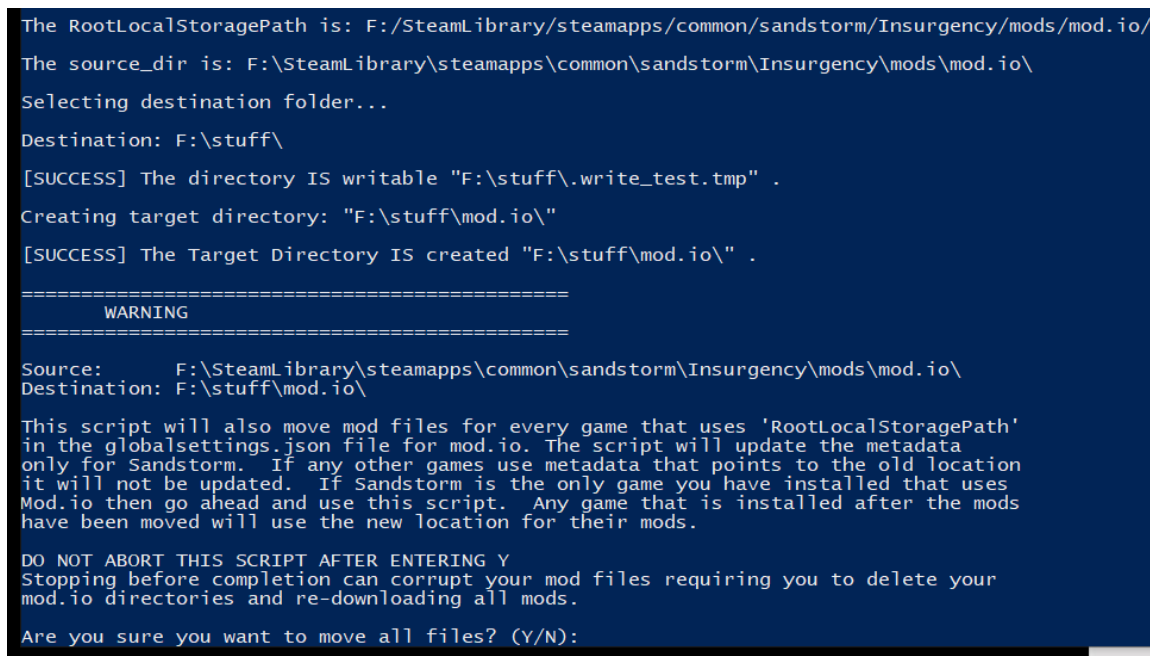
7. Move Mod.io Directory to new location

=====

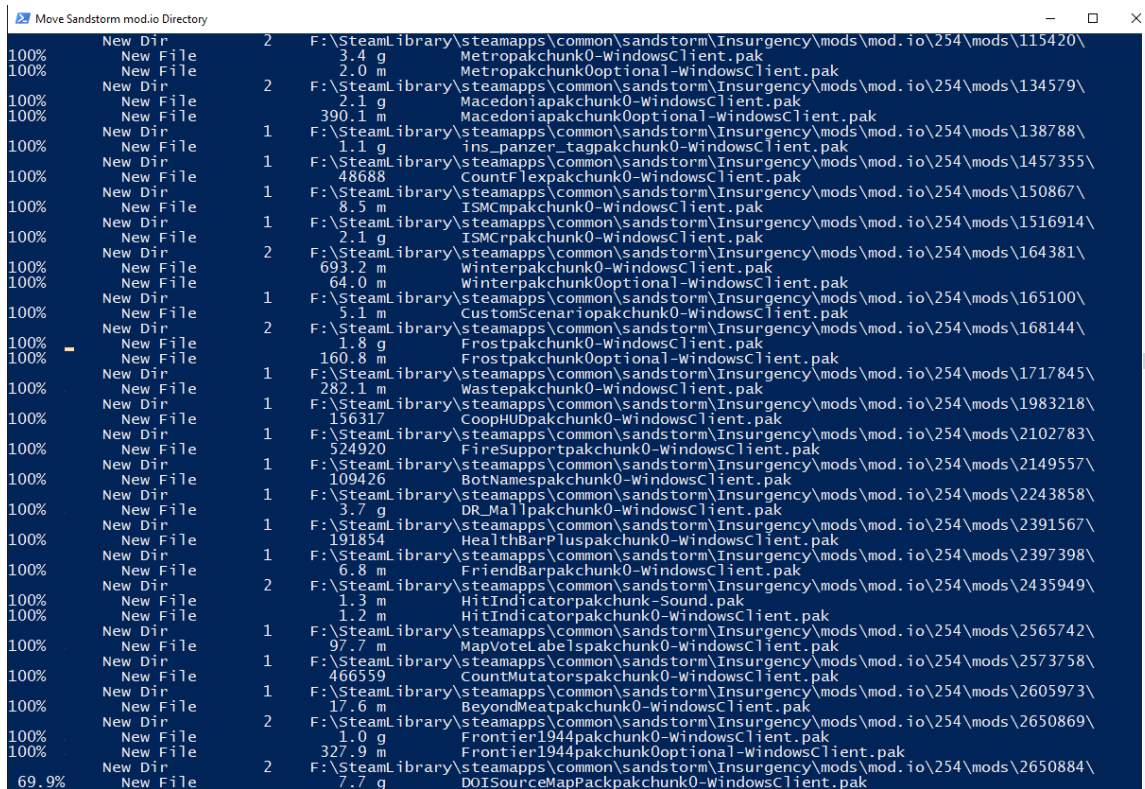
You can move your Sandstorm Mod directory to any location on your computer. You no longer need to keep your mods on the C:\Users\Public\mod.io location. This will free up space on your C: drive.



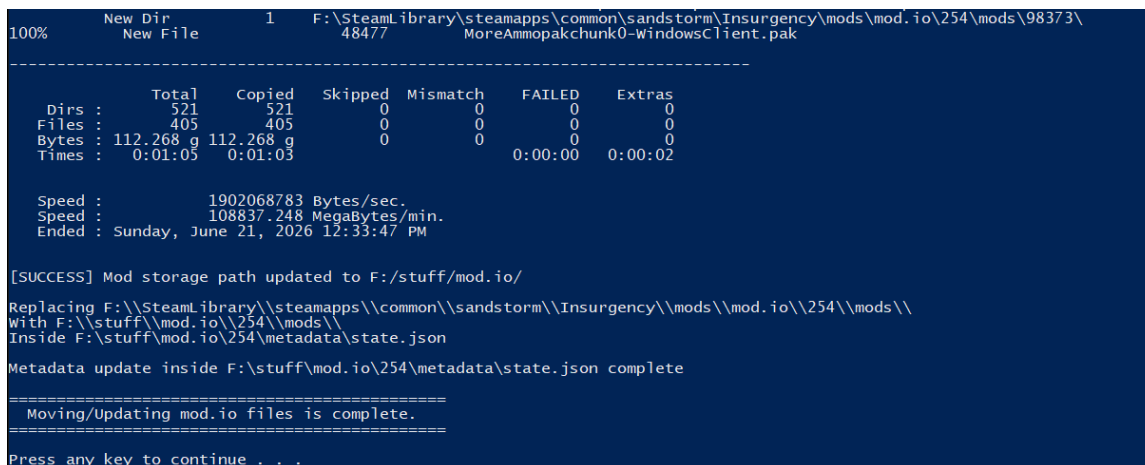
A GUI will open allowing you to select the location you would like the mods to be moved.



You will be given a last chance to back out of moving the files.



You will be shown the progress of moving every directory and file.



When finished you will be shown how many directories and files have been moved and if there were any errors.

8. Exit

=====

Kind of self explanatory. :)

Items of Note

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Server Operators

When you run this script to add newly subscribed mods to your server you may have to start the server twice. I have seen this happen when a server downloads new subscriptions on its own. It will sometimes stop booting while indicating the modlist is 0. When you restart the server it loads just fine. I have compared the state.json before and after with no change in it. I have no idea why it happens and it doesn't happen all of the time.

Players

If you join a server and it seems to hang on Downloading server mods... you can click the CANCEL button and then exit the game back to desktop. Then run this script as you have been subscribed to all of that server's mods and maps on mod.io. Running this script will download and install all of those mods to your game. When you load the game again it should go right into the server. You may get a short Downloading Server Mods... screen but just wait and you will get right in. After that you shouldn't see that screen again.

Final Note

The GUIs for Forced and Unsubscribed are now Alphabetized in ascending order